

FRANKLINE MISANGO OYOLO

+852 5104 2909 misango.me franklinemisango4@gmail.com github.com/FranklineMisango [in linkedin.com/in/misango](https://linkedin.com/in/misango)

SUMMARY

An practicing Quantitative Developer utilizing C++ and Python with applied research experience in equities, futures, and options. I am seeking a Market internship in foreign exchange, Derivatives, and Cryptocurrencies, with a focus on innovative pricing, advisory, and client-centric solutions.

EDUCATION

The University of Hong Kong

Bachelor of Engineering in Computer Science and Electrical Engineering— GPA: 3.73/4.3

Expected Dec 2027

Hong Kong SAR

PROFESSIONAL EXPERIENCE

Quantitative Developer

Aug 2025 – Present

Arithmax Research

United States

- Designs ultra low-latency medium frequency trading infrastructure in C++ with sub-millisecond event precision and high system uptime.
- Builds high-throughput market-making infrastructure in Rust and C# to run advanced mathematical model with jump-diffusion dynamics
- Engineers High-level research notebooks using Python with 99% vectorization efficiency using reducing strategy simulations time by 40%
- Programs simulations for VHDL/Verilog FPGAs and Assembly/C Operating System with real-time market data feeds for system migrations.
- Develops a suite of trading strategies across equities, options, crypto, and forex, helping traders generate at least 50% annualized returns.
- Leads American Markets strategies using GARCH-based volatility forecasting, forward rate bias correction, and risk-adjusted allocation
- Integrates alternative data sources like satellite imagery, social sentiment, and weather into factor models, enhancing regime-shifts robustness

Applied Machine Learning Researcher

Jul 2025 – Aug 2025

ETH Zurich & EPFL

Switzerland

- Published to prestigious ACM SIGSPATIAL and NeurIPS, proposing a customized DCRNN for demand forecasting of fiscal transactions
- Engineered a multi-scale feature integration pipeline using Python to extract demographic and credit transactions at a 1000m radius/station
- Formulated a sequence-to-sequence learning task using PyTorch DCGRU cells; extending framework for multimodal mobility EU networks
- Achieved 27% RMSE reduction over ST-GCN baselines with 13.4% gains from OSM integration; providing ablation studies for city planners

Technology Analyst

Jan 2025 – Jun 2025

BASF SE

Hong Kong SAR

- Architected a Retrieval-Augmented Generation pipeline using Python, GPT-5 and Azure Cognitive Search, achieving 93% query accuracy
- Developed a multi-agent system using AutoGen Framework with Python for SAP TPM integration, reducing shipment cost analysis to 38s
- Containerized Python analytics tools with Docker and deployed on Azure Kubernetes Service, achieving 99.95% uptime via HPAv2 autoscaling
- Optimized PySpark ETL pipelines on Databricks, processing 18TB/week of inventory and order data with 92% parallelization efficiency
- Implemented a route optimization using the DEAP algorithm in Python, reducing transportation costs by 31.4% EU logistics network

Applied Data Scientist

Jun 2024 – Aug 2024

Manulife Financial Corporation

Hong Kong SAR

- Built a hybrid contextual generation pipeline using Langchain, Python and GPT-4 Turbo, achieving 92.4% purview assets description generation
- Implemented dynamic few-shot learning with Azure Cognitive Search for 1.2M indexed documents, reducing metadata remediation effort
- Developed an automated GPT-Lint test suite using PyTest and Azure ML, improving compliance with 23 global data standards (GDPR, CCPA)
- Reduced LLM hallucination from 12.7% to 3.8% through weekly prompt toxicity scans using OWASP LLM top aligned against SWIFT taxonomy
- Architected test scenarios for generative AI features on Azure, stress-testing RAG pipelines at 12k QPS; identifying critical path bugs

ENGINEERING PROJECTS

FeenQR – Autonomous Quant Research Platform

C#, Python, HTML, CSS – 2025–Present

- Built an autonomous research platform using Microsoft Semantic Kernel and GPT-4 for multimodal signal generation
- Designed modular agents for market analysis and statistical pattern detection, enabling real-time alpha discovery and strategy orchestration.

TradeKernel – Nanosecond HFT Operating System

C++, Assembly – 2025–Present

- Engineered a bare-metal deterministic OS with 850ns event loop precision and 12.8M msg/sec throughput for ultra-low latency trading.
- Implemented AVX-512 kernel bypass, NUMA-aware memory allocator, and 4MB hugepages, improving real-time market data ingestion.

TurboBook – Order Book Imbalance Exploitation Engine

Rust, C# – 2025–Present

- Developed a multi-symbol market-making engine using broker WebSockets and real-time order book analytics to detect price inefficiencies
- Refactored C infrastructure for high-throughput execution and integrated multi-symbol quotations for latency-sensitive environments.

VeriTrade – FPGA Trading Simulation Environment

Verilog, VHDL, C++ – 2025–Present

- Designed a hardware-accurate simulation environment for FPGA-based trading accelerators, supporting both Icarus Verilog and Verilator.
- Built C++ testbenches and Dockerized simulation pipelines to validate latency-critical trading logic under real-time market conditions.

LANGUAGES, INTERESTS & ACTIVITIES

Languages: German (B1 Fluency), French (A2 Fluency), Mandarin (Beginner)

Interests: Chess (Blitz 1800, Rapid 1500), Leetcode (top 5%), Cycling, International travel, Reading on Financial Markets and Global Politics

Campus Leadership: HKU Trading Group TMT Software Developer, HKU Quantitative Researcher for Analysis of Algorithms

Awards: Amazon Web Services ABW Grant (2024, 2025), Python Software Foundation Grant (2024)

Certifications: AWS Machine Learning Specialty, AWS Solutions Architect Professional, Financial Modelling and Valuation Analyst (CFI)

TECHNICAL SKILLS

Programming: C++(Boost/STL), KDB+/q, C#, Python, Java (Springboot/Microservices)

Quantitative: Options pricing (BS/Heston), Order Book Dynamics, Volume Profile Analysis, FIX, Risk Management, PDE Solvers (Crank-Nicolson)

Business Tools: Alteryx, KNIME, SAP, Microsoft Power Apps, Databricks, Azure Data Factory

ML/AI: TensorFlow, PyTorch, XGBoost, CuDA, Natural language processing